Null Hypothesis: There is no significant difference between the sample average and the company's claim, i.e., $H_0 : \mu = 9.5$

Alternative Hypothesis: There is significant difference between the sample average and the company's claim, i.e., $H_1 : \mu \neq 9.5$ (two tailed test)

The level of significance $\alpha = 5\% = 0.05$

Applying the test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1);$$

$$Z = \frac{10 - 9.5}{\frac{3.5}{\sqrt{50}}} \sim N(0,1) = \frac{0.5}{0.495} = 1.01$$

Thus the calculated value 1.01 and the significant value or table value $Z_{\alpha/2} = 1.96$

Comparing the calculated and table value ,Here $Z < Z_{\alpha/2}$ i.e., 1.01<1.96.

Inference:Since the calculated value is less than table value i.e., $Z < Z_{\alpha/2}$ at 5% level of sinificance, the null hypothesis H_0 is accepted. Hence we conclude that the company's claim that the new car petrol consumption is 9.5 km per litre is acceptable.

Example 8.15

A manufacturer of ball pens claims that a certain pen he manufactures has a mean writing life of 400 pages with a standard deviation of 20 pages. A purchasing agent selects a sample of 100 pens and puts them for test. The mean writing life for the sample was 390 pages. Should the purchasing agent reject the manufactures claim at 1% level?

Solution:

Sample size n =100, Sample mean $\bar{x}$ = 390 pages, Population mean μ = 400 pages

Population SD σ = 20 pages

The sample is a large sample and so we apply Z -test

Null Hypothesis: There is no significant difference between the sample mean and the population mean of writing life of pen he manufactures, i.e., $H_0 : \mu = 400$

Alternative Hypothesis: There is significant difference between the sample mean and the population mean of writing life of pen he manufactures, i.e., $H_1 : \mu \neq 400$ (two tailed test)

The level of significance α = 1% = 0.01

Applying the test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1);$$

$$Z = \frac{390 - 400}{\frac{20}{\sqrt{100}}} = \frac{-10}{2} = -5, \ \therefore |Z| = 5$$

Thus the calculated value $|Z|$ = 5 and the significant value or table value $Z_{\alpha/2}$ = 2.58

Comparing the calculated and table values, we found $Z > Z_{\alpha/2}$ i.e., 5 > 2.58

Inference: Since the calculated value is greater than table value i.e., $Z > Z_{\alpha/2}$ at 1% level of significance, the null hypothesis is rejected and Therefore we concluded that $\mu \neq 400$ and the manufacturer's claim is rejected at 1% level of significance.

Example 8.16

(i) A sample of 900 members has a mean 3.4 cm and SD 2.61 cm. Is the sample taken from a large population with mean 3.25 cm. and SD 2.62 cm? (95% confidence limit)

(ii) If the population is normal and its mean is unknown, find the 95% and 98% confidence limits of true mean.

Solution:

(i) Given:

Sample size n =900, Sample mean $\bar{x}$ = 3.4 cm, Sample SD s = 2.61 cm.

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